TFB comprehensive audit — 22 Oct 2026 (one-time)

Audit window: 21 Oct 2026 21:00 UTC – 22 Oct 2026 (Asia/Riyadh 22 Oct 00:00 onward). Base commit: `264d3a9` (main). Live Render deploy: `55b47f2` (one commit behind main). Follows the 21 Oct audit (PR #809, safe subset merged in #810). This run carries the remaining learning-metric fix from #809 (with its review defect corrected), fixes three portfolio-math defects, and records the operational state.

Status labels: FOUND · PREPARED · APPLIED · TESTED · VERIFIED IN PRODUCTION · BLOCKED · ROLLED BACK

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1. الملخص التنفيذي (عربي)

الوضع الحالي: خدمة Render تعمل وتستجيب (جميع نقاط الفحص 200 خلال 0.3–0.7 ثانية) وتشغّل الإصدار `55b47f2`. مزامنة بيانات الأسواق نفسها ناجحة (Market_Leaders وGlobal_Markets بـ6,611 صفاً وMutual_Funds بـ2,474 صفاً كُتبت بلا أخطاء). لكن **صفحات القرار غير محدَّثة**: لوحة Top_10 عالقة على حالة "running..." منذ 21 أكتوبر 19:40 بتوقيت الرياض، وPortfolio_Decision لم تُحدَّث منذ 08:43 صباحاً بينما تحدّثت My_Portfolio الساعة 20:21. لذلك تفشل حواجز التحقق (وهي تعمل بشكل صحيح)، ولم تنجح أي مزامنة مجدولة كاملة في 22 من آخر 22 تشغيلاً منذ 14 أكتوبر.

ما تم إنجازه في هذا التدقيق (مُطبَّق ومُختبَر في فرع، لم يُنشر بعد): 1. أوزان المحفظة: سهم مملوك بلا سعر كان يُستبعد من المقام فتتضخم أوزان باقي الأسهم وقد تظهر توصية REDUCE خاطئة. الآن تُترك الأوزان فارغة للمحفظة كلها مع حالة `PORTFOLIO_VALUE_INCOMPLETE`. 2. أسعار الصرف: مفتاح `GBp` (بنس) كان يستبدل سعر الجنيه `GBP` فيُقيَّم المركز أقل بـ100 إلى 10,000 مرة. تم الإصلاح. 3. أوامر الخروج: كمية كسرية مثل 10.6 سهم كانت تقترح بيع 11 سهماً. الآن لا يتجاوز البيع ما هو مملوك. 4. مقاييس التعلّم: توصية بيع ارتفع سعرها كانت تُحسب "ربحاً". الآن تُقيَّم كل توصية حسب اتجاهها، وصُحِّحت نسبة Sharpe (كانت 13.42 بدلاً من 2.54 على المثال)، ويقارن معدل الثقة المعلن بالسجلات المحسومة نفسها.

المشكلة الرئيسية غير المحلولة: تشغيل `refreshDecisionTop10` في Apps Script لا يكتمل، ولا يُعاد تحديث `Portfolio_Decision` بعد المزامنة. مشروع Apps Script لم يكن متاحاً لي (BLOCKED)، فالإصلاح يتطلب إجراءً من المالك.

جاهزية التوصيات: غير جاهزة للاستخدام حالياً. صفحات القرار قديمة ومختلطة المصادر، والحواجز تمنع نشرها عن حق. لا تعتمد على Top_10 أو Portfolio_Decision حتى يكتمل تحديث ناجح وتنجح بوابة Decision Surface Freshness.

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2. Access and baseline

Resource	Access	Notes
GitHub repo	Full (read/write via branch + PR)	723 tracked files, ~323k lines of py/gs/js; 38 workflows; 21 open PRs
GitHub Actions	Read logs	No re-runs triggered
Render service `srv-d4hnir15pdvs739bqe1g`	Read (events, logs, metrics)	Auto-deploy **off**; deploys are manual. No deploy triggered by this audit
Google Sheet `1QaOSNH…`	**Partial** — metadata + first 5 tabs via Drive text export	Decision tabs (Portfolio_Decision, Top_10_Investments, v2, My_Portfolio, _Status) **BLOCKED**: the export truncates before them; xlsx export not supported by the connector
Apps Script project `1IvEj8…`	**BLOCKED** — "Requested entity was not found" (container-bound script, not visible to Drive)	Repo copy in `apps_script/` could not be compared with the deployed script
Dashboard site	Not inspected	No authenticated browser in this session — **BLOCKED**
IBKR	Not used (per instructions)	Broker reconciliation **not verified**

Baseline metrics (sources and denominators)

Metric	Value	Window / source
Live probes `/health /healthz /readyz /livez /ping /meta /v1/meta`	8/8 HTTP 200, 0.30–0.70 s	22 Oct 00:28 Riyadh, single probe each

Metric	Value	Window / source
Live commit	`55b47f2` = `/meta deploy.render_git_commit`	main is `264d3a9` (#813, Athens venue calendar) — not deployed
HTTP status, 7 days	200: 13,555 · 499: 212 · 401: 84 · 502: 6 (0.04%)	Render metrics 14–21 Oct
HTTP status, 24 h	200: 2,456 · 499: 50 · 401: 75 · 5xx: 0	Render metrics
Worker SIGSEGV (`code 139`)	5 in 7 days	18 Oct 08:55Z, 19 Oct 23:48Z, 20 Oct 09:30Z ×2, 20 Oct 23:17Z
Memory per instance	climbs to 3.8–4.2 GB of 4.29 GB before each deploy resets it	current instance 0.73 → 3.80 GB in 7 h
Latency p50/p95	**not available** (Render `http_latency` returned empty)	—
Scheduled Dashboard Advanced Sync	**0 of 22** succeeded	scheduled runs 14–21 Oct
Decision Surface Freshness	**0 of 10** non-skipped runs succeeded	last 100 runs (90 skipped by filter)
Data-sync step inside the failed 21 Oct 17:20Z run	3/3 legs `failed=0`; rows written: Global_Markets 6,611, Mutual_Funds 2,474	job logs
Offline test suite at base	5,438 passed · 37 skipped · 0 failed	local venv with requirements.txt pins, Python 3.11

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3. Architecture findings (Providers → Python/API → Apps Script → Sheets → dashboard)

1. **Providers → Render (FastAPI).** Healthy. Error logs are dominated by Yahoo throttling (`yfinance circuit OPEN`, 404 "Quote not found" for real tickers such as BK and SPLG). The breaker and EODHD fallback work as designed: EODHD `f429=0`, no 5xx in 24 h. FOUND: `[FRESHNESS v1.27.0] venue calendar unavailable` for `.AT .VN .NS .AE .QA .BH`. `.AT` is fixed on main (#813) but not deployed. The rest have no calendar in exchange_calendars 4.13.2 and stay fail-closed by design.
2. **Render runtime.** FOUND (High): memory grows steadily on every instance until a deploy restarts it. Five native segfaults happened in 7 days, two of them at 95% of the memory limit. Frequent manual deploys are currently what prevents OOM kills. Root cause is not located (no Python traceback; likely a native library under memory pressure). Backlog B2.
3. **GitHub Actions sync → Sheets.** The market-data legs write cleanly. Every leg then fails `Validate Dashboard` on `decision.header_found`, because the Top_10 cockpit is still `running…`, 51–108 min after its stamp. The guard is correct and was added in #810.
4. **Apps Script decision layer (FOUND, Critical, BLOCKED for repair).** `refreshDecisionTop10` does not finish, and Portfolio_Decision is not re-run after syncs. v1.13.10 writes FAILED when the refresh throws, but the live status is RUNNING. That means either the 6-minute Apps Script timeout (which skips the catch) or v1.13.10 not being installed. I could not tell which, because the script project is inaccessible.
5. **Freshness gate.** 12 FAILs on 21 Oct 18:30Z: `PF_RUN_STALE`, `PF_OLDER_THAN_SOURCE`, `T10_STATUS_NOT_OK`, `T10_OLDER_THAN_SOURCE` ×4, `T10_POOL_COUNT_MISSING` ×4 and `T10_POOL_RECONCILIATION_UNPROVEN`. All follow from item 4. Loosening the gate would publish stale decisions, so it was left unchanged.
6. **Version reporting.** FOUND (Low): `/meta app_version` reads 8.14.5 while `entry_version` reads 8.14.6. `main.py:1034` takes `APP_VERSION` from the environment, so the Render env var is stale. This is a config change for the owner.
7. **FX configuration.** FOUND (Medium): Render logs `PF-WEIGHT-FX operator FX rates are 61 days old (TFB_PF_FX_AS_OF), limit 30`. The operator-supplied non-peg rates used for My_Portfolio SAR weights need a refresh, which is a config change for the owner.

Page-by-page assessment

Page	Producer	State at audit	Assessment
Market_Leaders / Global_Markets / Commodities_FX / Mutual_Funds	`run_dashboard_sync.py` (Actions)	Synced 21 Oct, `failed=0`	Data refresh working; Global_Markets meets 6,512 floor (6,611)
My_Portfolio	sync + `10_My_Portfolio.gs`	Refreshed 21 Oct 20:21 Riyadh	Weights affected by defects D1/D2 (fixed in this PR, pending deploy)
Portfolio_Decision	Apps Script	Last run 21 Oct 08:43 Riyadh	**Stale** — older than its source; do not act on it

Page	Producer	State at audit	Assessment
Top_10_Investments	Apps Script `16_Decision_Top10.gs`	`running…` since 19:40 Riyadh, no grid	**Unavailable** — no qualifying opportunities can be confirmed
v2 decision pages	Apps Script	**BLOCKED** (not readable)	—
`_Data_Audit`, `Diagnostic_Report`	legacy	Last written Jun 2026	Abandoned; their timestamps can mislead readers. Backlog B7
Dashboard site	external	**BLOCKED**	Not inspected

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4. Implementation register

Branch `claude/vibrant-bell-3xt7y7`, one PR (link in the PR list). Nothing is deployed: Render auto-deploy is off and deploying is the owner's call.

ID	Defect and root cause	Files	Status
D1	**Unpriced holding inflated every other weight.** `_compute_portfolio_fields` summed only rows with a `position_value`, so a held row with no price left the denominator. Example: 2222.SR 100@25 plus 1120.SR 100@None gave 2222.SR weight 100%, gap −50 and a drift REDUCE, all under status `OK:SAR`. Fix: such a book is blanked like `PORTFOLIO_FX_INCOMPLETE`, with status `PORTFOLIO_VALUE_INCOMPLETE: <symbols>` and no drift ADD/REDUCE. A carried `position_value` still counts.	`core/data_engine_v2.py` 5.155.0 → 5.156.0	APPLIED · TESTED
D2	**Pence FX key overwrote the pounds rate.** `_pf_fx_table` upper-cased every key, so `GBp:0.048` replaced `GBP:4.8`. Fix: `_pf_fx_key` files GBp/ZAc/ILa under GBX/ZAC/ILA, which the per-row lookup already reads as SAR per subunit.	`core/data_engine_v2.py`	APPLIED · TESTED
D3	**EXIT oversold fractional holdings.** `int(round(10.6))` = 11. Fix: floor to whole shares, and price per share at value ÷ true quantity, so the remainder stays visible in post-trade weight. The delta stays a signed integer (the 10_My_Portfolio contract).	`core/analysis/portfolio_actions.py` 1.16.0 → 1.16.1	APPLIED · TESTED
D4	**Learning metrics scored the price, not the call** (carried from #809). A SELL that rose counted as a win. Hit rate divided by all records. Sharpe treated percent horizon returns as daily fractions (13.42 vs 2.54). Fix: direction-aware outcomes (HOLD neutral), decided-only rates, and per-horizon Sharpe/Sortino. Stored Outcome/ROI cells are unchanged.	`scripts/track_performance.py` 6.45.0 → 6.46.0	APPLIED · TESTED
D4b	**#809 review defect.** The calibration "claimed" rate averaged reliability over all records (HOLD and breakeven included), while the realized rate used decided records only. Fix: claimed rate over the same decided records; `mean_claimed_reliability` keeps the all-n average.	`scripts/track_performance.py`	APPLIED · TESTED
—	Manifest pins (engine 5.156.0, portfolio actions 1.16.1, tracker 6.46.0); CI enrolment	`scripts/verify_deployment.py`, `.github/workflows/ci.yml` 1.1.30, `tests/test_core_repair_tiny_quotes.py`	APPLIED

Not carried from #809: the `run_shadow_scorer` 1.9.4 turnover/cost change. It is a methodology change with no switch or version-break marker, and it would restart the S-1 evaluation window. It needs an owner decision.

Tests. New: `tests/test_pf_weight_value_complete.py` (14 cases) and `tests/test_learning_metric_fixes.py` (30 cases). Golden negative on base `264d3a9`: 39 of 44 fail and all 44 pass with the fix. The 5 that pass on base are the no-regression cases. Both suites also pass under the lean CI lane dependencies (numpy + pytest). Whole suite on the branch: **5,482 passed, 37 skipped, 0 failed** (base 5,438 + 44 new).

Rollback. Revert the PR's commit. There are no flags, migrations, schema or sheet-contract changes. Behaviour visible after deploy: - A My_Portfolio book with any unpriced holding shows blank weights and gaps with `PORTFOLIO_VALUE_INCOMPLETE`, not

inflated numbers. - Tracker win rates score SELL-family calls by direction. Sharpe/Sortino print per horizon. - `Calibration claimed` rates move to the decided-record basis.

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5. Prioritised remaining backlog

#	Priority	Item	Dependency	Acceptance criteria	Effort
B1	Critical	Make `refreshDecisionTop10` finish. **Update (22 Oct):** #815, merged to main while this audit ran, ships `16_Decision_Top10.gs` v1.13.11. It writes the cockpit formats in bulk so the refresh finishes inside the 6-minute Apps Script limit, which addresses the root cause. Remaining owner steps: paste v1.13.11 into the bound Apps Script project, then re-run `refreshDecisionTop10` and `refreshPortfolioDecision` after a sync	Apps Script access (owner)	Cockpit status OK with `pool_reconciliation_v1=` receipt; Decision Surface Freshness passes; next scheduled Advanced Sync green	30 min (owner)
B2	High	Render memory growth and SIGSEGV workers	Render access, profiling	24 h without memory above 80% or code 139; consider `--max-requests` worker recycling (no cost increase)	0.5–1 day
B3	High	Deploy main (`264d3a9`) plus this PR after merge	Owner merge + manual deploy	`/meta render_git_commit` equals the merge SHA; `verify_deployment --strict` CLEAN	15 min
B4	Medium	Refresh `TFB_PF_FX_RATES` / `TFB_PF_FX_AS_OF` (61 days old) and correct the stale `APP_VERSION` env	Render env (owner)	No `PF-WEIGHT-FX … days old` log line; `app_version` = `entry_version`	10 min
B5	Medium	Decide on the shadow-scorer turnover/cost methodology (#809) with a version-break marker	Owner decision	S-1 window restart documented, or change gated	2 h
B6	Medium	Run the Top_10 cockpit check in one sync leg only, so a stuck cockpit doesn't turn three healthy data legs red	Owner decision (alerting change)	Data-leg failures reflect data problems only	1 h
B7	Low	Retire or re-date the `_Data_Audit` and `Diagnostic_Report` tabs (last written Jun 2026)	Sheet access	No stale diagnostic tabs visible to readers	1 h
B8	Low	Triage the 21 open PRs, several with overlapping version claims (engine 5.155.0 also claimed by #794)	Owner	Open PR count reduced; no duplicate version claims	2 h
B9	Low	Turn on the peg-band check (`TFB_OPP_FX_SANITY`, default off), so an inverted USD/SAR rate cannot be accepted	Owner decision	Inverted 0.2667 rate rejected in a test	1 h

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6. Before / after

Measure	Before (base `264d3a9`)	After (this branch)	Verified in production?
Unpriced-holding book: 2222.SR weight / action	100% / REDUCE under `OK:SAR`	blank / no drift action, `PORTFOLIO_VALUE_INCOMPLETE: 1120.SR`	No — awaits deploy
`GBP:4.8,GBp:0.048` → GBP table rate	0.048	4.8 (GBX 0.048)	No
EXIT of 10.6 shares → delta	−11	−10 (0.6 remains in post-trade weight)	No
Two SELL calls that rose: win rate	100%	0%	No
1M ROIs [2,−1,3,1]% → Sharpe	13.42	2.54	No
Scheduled Advanced Sync success	0/22	unchanged — blocked on B1	—

Measure	Before (base `264d3a9`)	After (this branch)	Verified in production?
Offline suite	5,438 passed / 37 skipped / 0 failed	5,482 passed / 37 skipped / 0 failed	n/a

Forecast performance: not reported. The tracker's direction-aware metrics only take effect after deploy, so no completed-horizon sample under the corrected definitions exists yet.

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7. Final verdict

- **Operationally healthy:** *Partially.* The API is up, serving the expected commit, and the market-data refresh works. But the decision layer has not produced a complete Top_10 / Portfolio_Decision refresh since 21 Oct morning. The Render workers leak memory and segfault several times a week.
- **Financially consistent:** *Not yet in production.* Three portfolio-math defects and four learning-metric definition defects are fixed and tested on the branch, but not deployed. FX operator rates are 61 days old. Broker reconciliation was not verified (IBKR deliberately not used; no statements supplied).
- **Sufficiently validated for recommendation use: No.** The decision surfaces are stale and the freshness gates correctly block them. Treat any current Top_10 / Portfolio_Decision content as not qualifying until B1 is resolved and the gate passes.